








# Early–late asymmetry of the muon density with the UMD

## Selection bias, core reconstruction, and how to recover it

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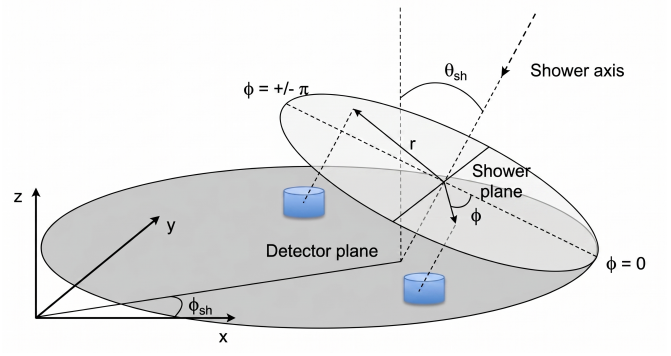
ITeDA (CNEA-CONICET-UNSAM) — FCEN-UBA

**UMD Foundations Meeting** — 4 November 2026

# The observable: first harmonic in the shower plane

- $\phi$: station azimuth in the **shower plane**.
- Vertical shower: circular symmetry.
- Inclined shower: two distinct regions.
  - **early** ($\phi \approx 0°$): reached first, through *less* atmosphere.
  - **late** ($\phi \approx \pm 180°$): reached later, through *more* atmosphere.
- $A_1 > 0$: early excess; $A_1 < 0$: late excess.
- Fitted on an **ensemble** of showers, per $(r, \theta)$ bin.

$$\rho(r, \phi) = \langle \rho(r) \rangle \, (1 + A_1 \cos \phi)$$



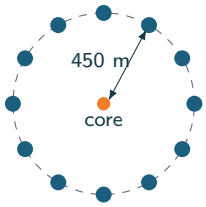
# The reference: Monte Carlo geometry

# Simulations and the Dense Ring

- Official CORSIKA + *Offline* production, with SD-750 and UMD:

| | |
|---|---|
| **Hadronic models** | SIBYLL 2.3e (main), QGSJET-III.01, EPOS-LHC-R |
| **Primaries** | p, He, O, Fe |
| **Energy** | $10^{17.5}$–$10^{18}$ eV |
| **Zenith angle** | $\theta \leq 65^\circ$ |

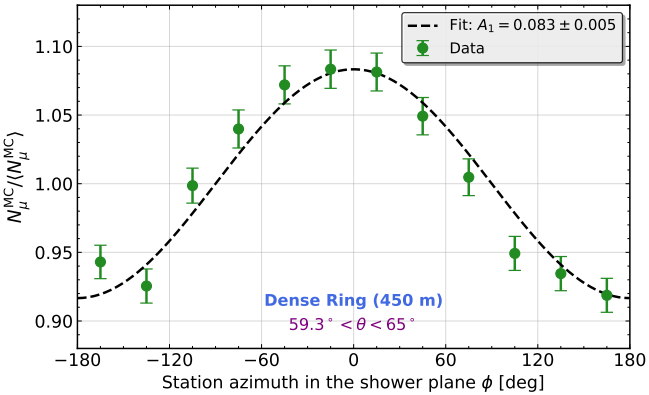
- **Dense Ring**: 12 virtual stations at $r = 450$ m from the **true** axis, every 30° in the shower plane. Same $r$ for all: the radial fall-off of the density cannot produce a modulation in $\phi$.




Shower plane.

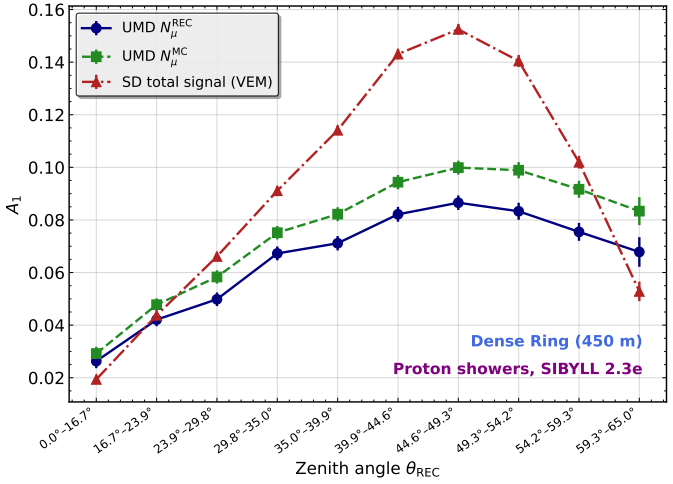
# A clear modulation on the Dense Ring






- Injected muon number $N_\mu^{MC}$, $59.3° < \theta < 65°$.
- Maximum in the **early** region ($\phi \approx 0$).
- Early–late difference $\approx 17\%$ peak to peak ($\pm 8\%$ around the mean).

# Comparison: SD vs. UMD on the Dense Ring

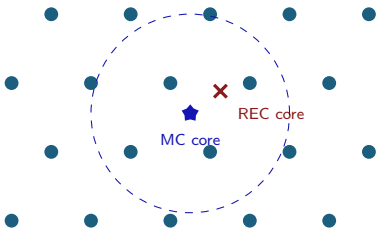



- UMD: $A_1 > 0$, grows with $\theta$ up to $\sim$ 45–50°.
- UMD: lower amplitude with reconstructed ($N_\mu^{\mathrm{REC}}$) than with injected ($N_\mu^{\mathrm{MC}}$) muons, same shape.
- SD total signal: larger than the UMD at $\sim$ 20–60° (EM-dominated); drops sharply above $\sim$ 50°.
- SD below the UMD only at $\theta \gtrsim 60°$.

# The Infill array: two geometries



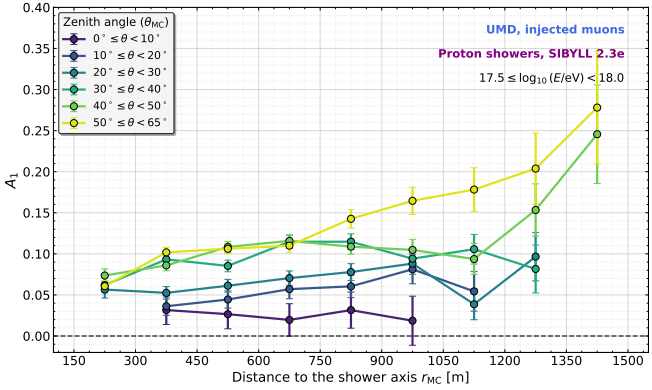
- Now the real Infill stations (750 m): the distance to the axis varies from station to station $\Rightarrow$ we can study $A_1(r)$.
- $r$ and $\phi$ of each station depend on where the shower axis is. Two geometries:
  - **MC**: true core and direction;
  - **REC**: reconstructed core and direction.
- The main difference between them is the **core position**.




Stations every 750 m. With another core, $r$ and $\phi$ of every station change.

# UMD radial dependence on the Infill (MC geometry)





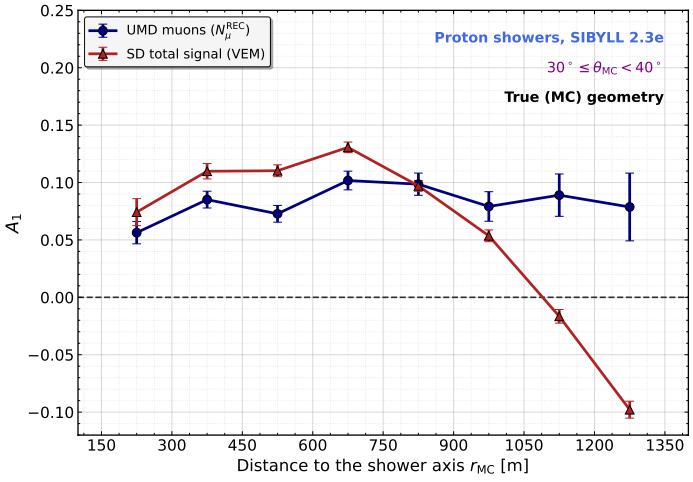

- **UMD only** (injected muons); $r, \phi$ from the **true** core and direction.
- Radial bins of $\Delta r = 150$ m; zenith bins of $10^\circ$.
- $\theta < 10^\circ$: compatible with zero.
- 10–$40^\circ$: grows with $r$, then **flattens**.
- $\theta \geq 40^\circ$: **grows with** $r$.

**This is our reference.**

# The SD inversion and the selection bias

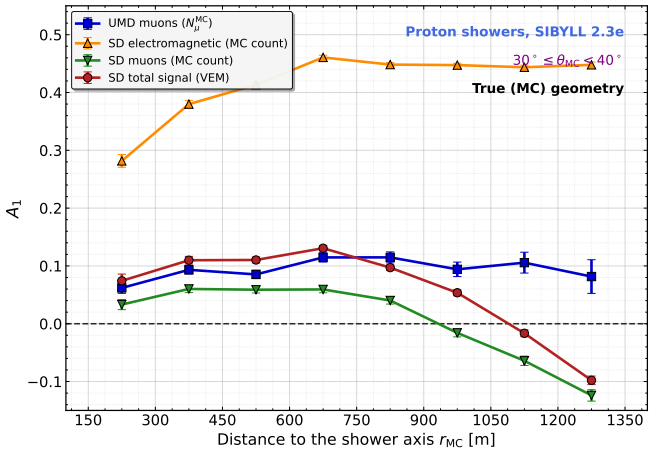
# The SD comparison: inversion far from the core






- We now compare the UMD and SD asymmetries in the same Infill showers.
- MC geometry, $30^\circ \leq \theta < 40^\circ$.
- UMD: positive at all $r$.
- SD total: a maximum, then negative far from the core.

# It is the SD muon component that inverts



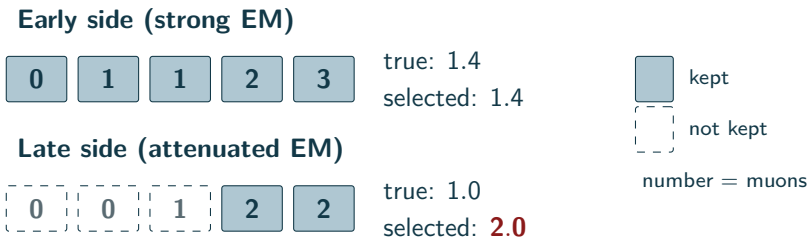

- MC particle counts in the tank, split by component.
- **EM**: positive at all $r$.
- **SD muons**: positive near the core, negative beyond $\sim$ 900 m.
- The SD total follows its muons at large $r$.

**Same quantity, opposite sign in UMD and SD.**

# Hypothesis: the station selection biases the average

HYPOTHESIS

A station enters the analysis only if it **belongs to the reconstructed event** (`HasStation`). If the probability that a station is kept depends on its own muon content, and this dependence is not the same on the early and late sides, the **regional averages are biased**.

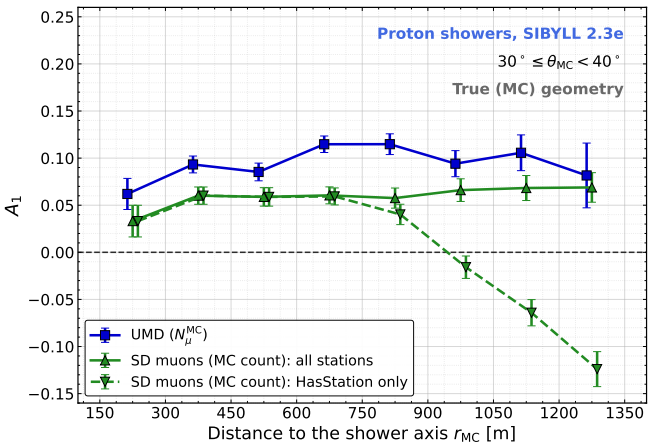



The sign can flip **without moving a single muon**.

Illustrative cartoon, not simulation output.

# Test 1: without HasStation, the inversion disappears



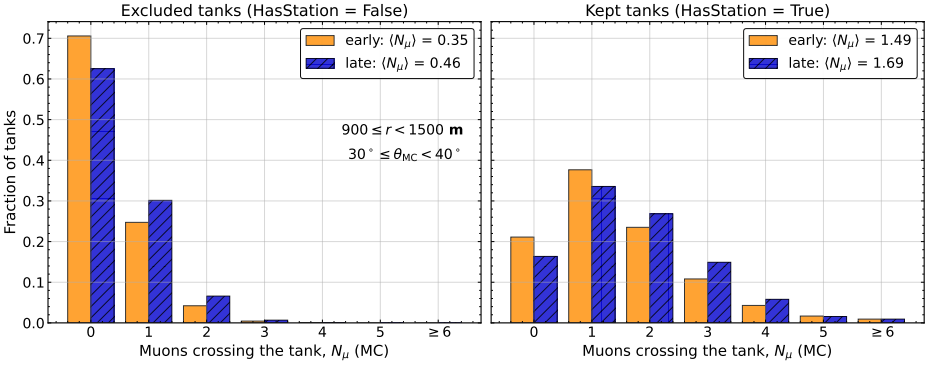



- Same MC counts, coordinates and fit: only **which stations** enter changes.
- All stations: SD muons **positive at all** $r$ ($+0.03$ to $+0.07$).
- Same sign and trend as the UMD, **but not the same values**.

# Test 2: the excluded tanks should hold very few muons



- If the inversion comes from the **trigger**, the tanks dropped by the reconstruction (`HasStation` = False) should have been crossed, on average, by at most one muon, too few for the station to trigger.
- Excluded tanks: $\leq 1\,\mu$ in 95% (early) and 93% (late) of the cases.



# Reconstructed geometry and the core position

# With REC geometry, the UMD asymmetry vanishes

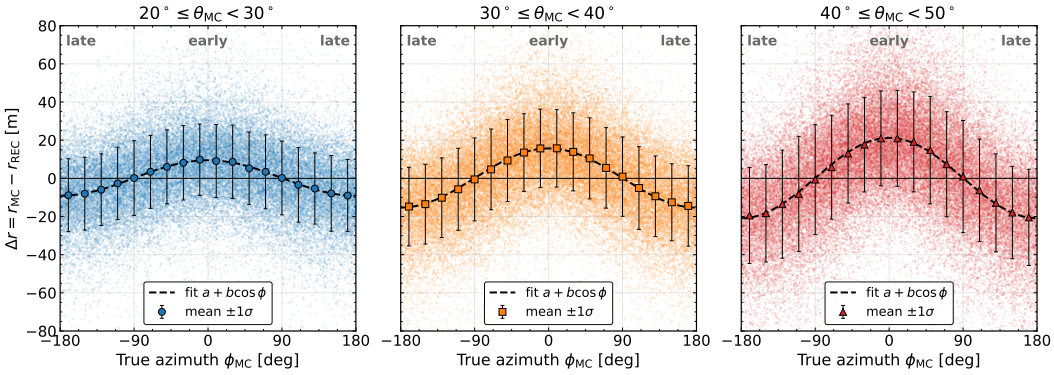
- Same UMD modules, same **injected** muon number, $450 \leq r < 600$ m.

| $\theta$ | $A_1$ Monte Carlo | $A_1$ Reconstructed |
|---|---|---|
| 20°–30° | 0.062 | 0.011 |
| 30°–40° | 0.086 | 0.002 |
| 40°–50° | 0.108 | −0.002 |

- With the **reconstructed core**, the asymmetry is lost almost completely.
- The effect comes from the **offset between the true and the reconstructed core positions**.

# The reconstructed core is shifted toward the early region






- $\Delta r = r_{\mathrm{MC}} - r_{\mathrm{REC}}$ vs. $\phi$ can be fitted by $a + b\cos\phi$: early stations are placed **closer** to the core, late ones **farther**, with $b \simeq 9$, 15, 21 m, increasing with $\theta$.
- Likely cause: the reconstruction LDF is **symmetric in** $\phi$; it absorbs the early excess by **moving the core toward the early side, where there is more signal**.

# A shifted core adds a $\cos\phi$ term to the LDF

- Approximating the LDF locally as $\rho \propto r^{-\beta}$ and correcting it for the core shift with an azimuthal dependence, $r \to r + b\cos\phi$:

$$\rho(r + b\cos\phi) \simeq \rho(r)\left[1 - \beta\,\frac{b}{r}\cos\phi\right] \;\Rightarrow\; \boxed{\Delta A_1 \simeq -\beta\,\frac{b}{r}}$$

| $\theta$ | $A_1$ (MC) | $A_1$ (REC) | $\Delta A_1$ | $A_1^{\mathrm{REC}} + \beta b/r$ |
|---|---|---|---|---|
| 20–30° | 0.062 | 0.011 | −0.051 | **0.062** |
| 30–40° | 0.086 | 0.002 | −0.084 | **0.083** |
| 40–50° | 0.108 | −0.002 | −0.110 | **0.100** |

- The shift accounts for **93–100%** of the loss; adding it back **recovers the MC reference asymmetry** to $\lesssim 0.01$.

# Recovering the asymmetry: what it needs, what is missing

## What the correction needs

- $b(\theta)$: mean core shift, early–late direction. Measured with MC truth.
- $\beta$: local slope of the muon LDF. From MC here; in principle measurable in data.
- Apply: move the core back by $b$, or add $\beta b/r$ to $A_1$.

## Established

The loss is **systematic**, not resolution, and predicted by $\beta b/r$ (450–600 m, 20–50°).

## Not done yet

- Move the core and recompute $A_1(r)$ at all $r$.
- Counting bias ($N_\mu^{\mathrm{REC}}$, 5–23%): separate effect.
- **For data** (no true core), $b$ must come from:
  - $b(\theta)$ parametrized on simulations and **validated on unseen ones** (other showers, primaries, models; tested so far: p, SIBYLL 2.3e);
  - or an LDF fit with an azimuthal term in the reconstruction.

# Conclusions

# Conclusions

1. In simulations the UMD muon density has a **positive early–late asymmetry**: it grows with $r$ for inclined showers and is roughly flat at intermediate angles.
2. The SD muon asymmetry inverts far from the core **only** when averaging over the stations kept by the reconstruction: the selection acts **station by station**, not shower by shower. Over all stations it is positive. The excluded tanks hold **mostly** $\leq 1$ muon: **consistent with** a trigger effect, not yet proven.
3. With the reconstructed geometry the UMD asymmetry nearly vanishes near the core: the reconstructed core is shifted toward the early region by $b \approx 9$–$21$ m.
4. This loss follows $\Delta A_1 \simeq -\beta\, b/r$; adding it back recovers the MC reference. Applying it to data requires $b$ **without MC truth**.

**Data–MC comparisons need the same station selection and the same geometry.**

# Next steps

- Parametrize the correction parameters vs. zenith angle: $b(\theta)$ and $\beta(\theta)$, at fixed energy ($17.5 \leq \lg E < 18.0$), and validate the recovery on independent simulations.
- Apply the full correction (core shift + LDF term) over the whole radial range and compare with the MC reference.
- Dependence on primary and hadronic model.
- Separate trigger, readout and reconstruction inside `HasStation`.
- Apply to field data.

# Thank you!

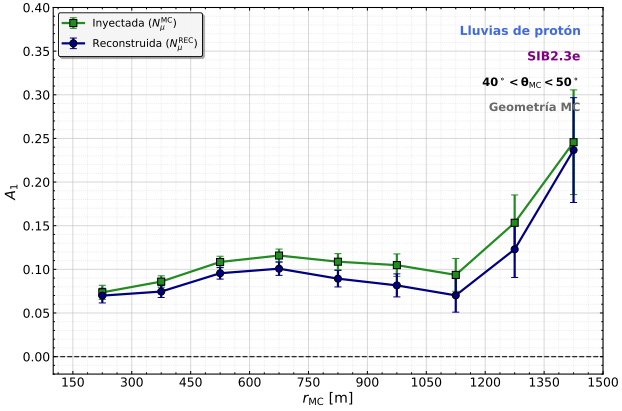
## Questions?



Image: A. Chantelauze, S. Staffi, L. Bret (Pierre Auger Observatory)

# Backup: muon-counting reconstruction (MC geometry)

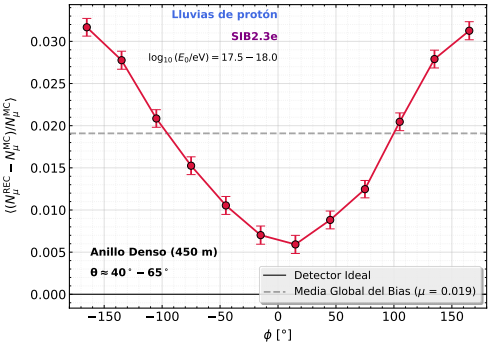


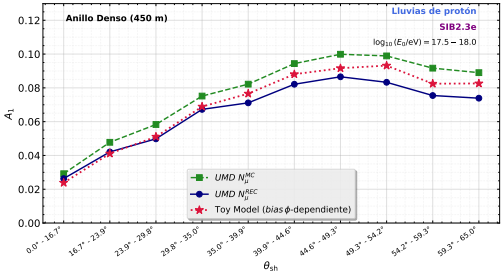



- 40°–50°, true core and direction.
- Reconstructed muon number lowers $A_1$ by 5–23%, same shape.
- A separate effect from the core shift.

# Backup: the counting bias depends on azimuth





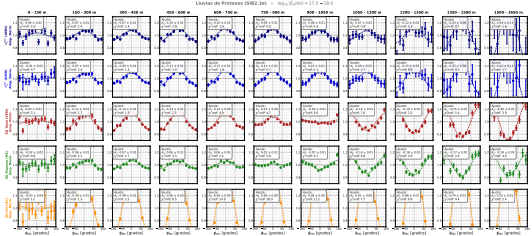




- Mean relative residual $(N^{\mathrm{REC}} - N^{\mathrm{MC}})/N^{\mathrm{MC}}$ is larger on the late side (fewer muons) $\Rightarrow$ lower $A_1$.
- Bootstrap with the per-region residuals reproduces $A_1^{\mathrm{REC}}$ up to $\theta \approx 35°$; above, half to two thirds of the loss. Remnant open.

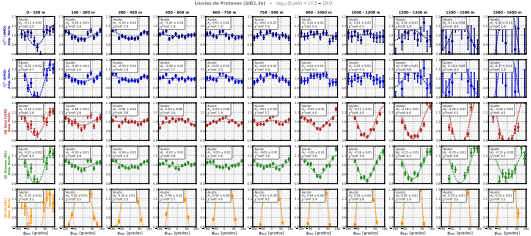
# Backup: fit grids, MC and REC geometry (40°–50°)



## MC geometry

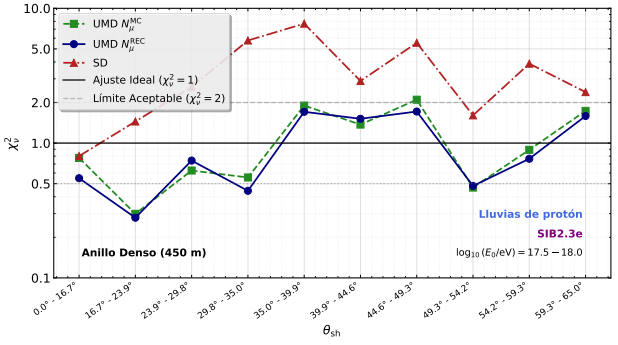



## REC geometry



# Backup: fit quality on the Dense Ring

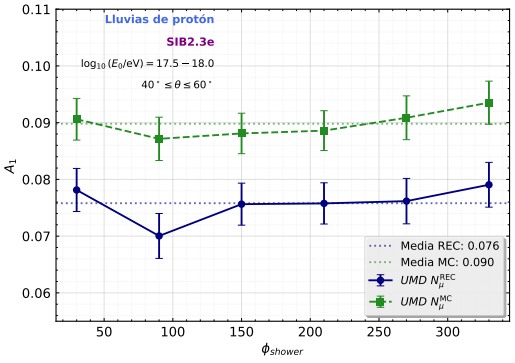




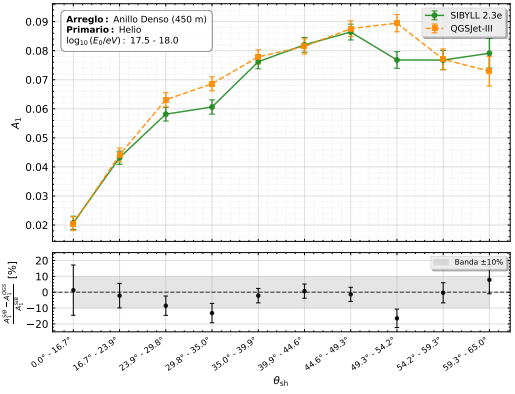

- UMD, MC and REC: $\chi^2_\nu \approx 1$.
- SD: $\chi^2_\nu$ up to $\sim 10$; with such small errors the fit is sensitive to higher harmonics and to correlations between stations of one shower.

# Backup: robustness

SIMULATIONS – DENSE RING

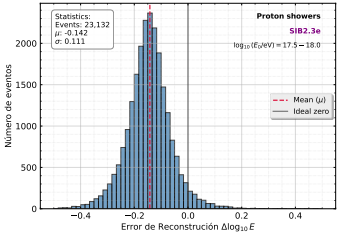


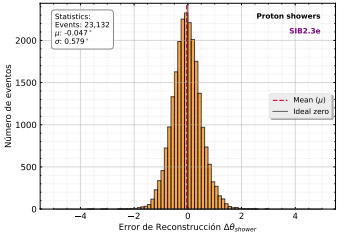

Independent of the shower arrival azimuth (40–60°).

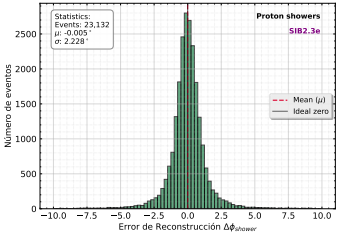



Helium: SIBYLL 2.3e vs. QGSJET-III within ±10% over most of the range.

# Backup: reconstruction performance



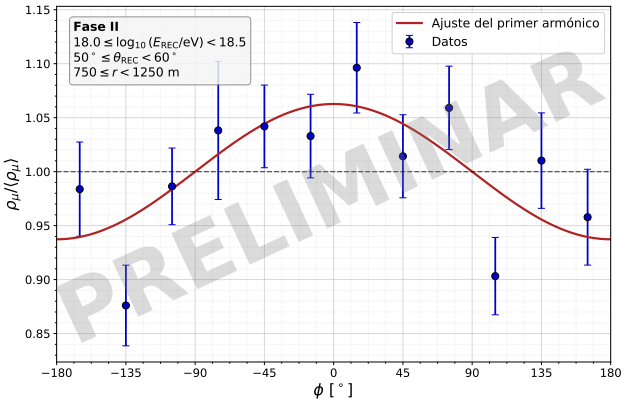

- Energy: $\mu \approx -0.142$, $\sigma \approx 0.111$ in $\Delta \log_{10} E$.
- $\theta$: bias $-0.05°$, $\sigma = 0.58°$. $\phi$: bias $-0.005°$, $\sigma = 2.2°$.

# Backup: why the core shift does not affect the Dense Ring

- The ring stations are placed during the simulation at 450 m from the **true** axis: the shifted core does not make the ring eccentric.
- It only rotates the azimuth assigned to each station, by $\sim b/r \approx 2^\circ$, much less than the $30^\circ$ spacing.
- It affects $N_\mu^{\mathrm{MC}}$ and $N_\mu^{\mathrm{REC}}$ in the same way, so it cannot explain the MC–REC gap on the ring (that gap is the counting bias).

# Backup: field data (preliminary)

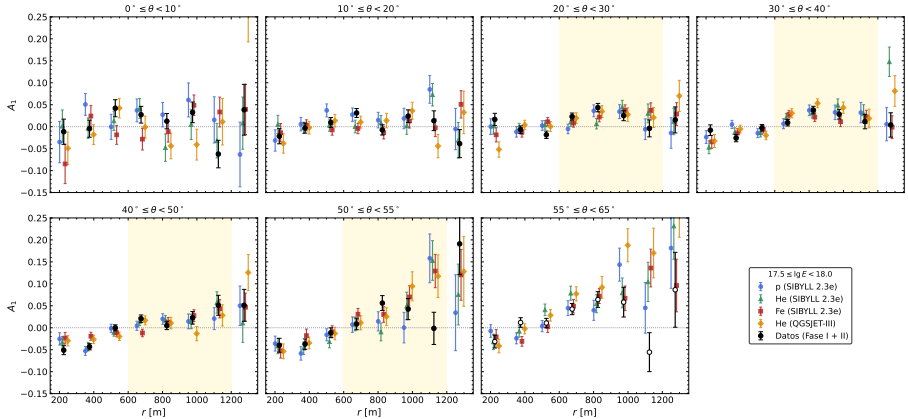
FIELD DATA – PRELIMINARY




- UMD muon density vs. $\phi$, normalized to its mean; Phase II, cuts in the box.
- Maximum at $\phi \approx 0$: **early excess**, as in the simulations.
- **Preliminary**: no core-shift correction, no systematics, no MC above $10^{18}$ eV for this bin.

# Backup: $A_1(r)$ in field data vs. simulations

FIELD DATA – PRELIMINARY




UMD Infill, Phase I + Phase II, 6T5, $17.5 \leq \lg(E/\mathrm{eV}) < 18.0$, REC geometry in data and simulations.